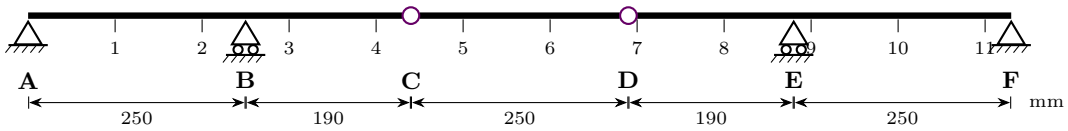
Predict – scan – check

Lab S2 Influence Lines: in-class activity (15 minutes, in pairs)

Name(s): _____ Group: _____

Write your prediction *first*, in ink. Then scan the code (it opens the *Explorer* tab of the Lab S2 course page), do the same task in *Part 1 · Explorer tasks*, and record what you read. A wrong prediction with a good explanation of what you learned earns full credit.

Lab S2 page, Explorer tab



1. **Read the reactions.** A unit load stands at halt 6, on the suspended span. Predict which of the six reactions are non-zero, and their signs.

   Explorer (*Quick fill: Unit load*, halt 6): $R_C =$ _____ $R_E =$ _____

2. **Open an explanation.** Tap the R_E box. $\eta_D =$ _____. In one line: why is $\eta_E = 1.76\,\eta_D$?

3. **Set your own loads.** $P_1 = 3\,\text{kg}$, $P_2 = 2\,\text{kg}$, $P_3 = 0$, spacing 140 mm, P_1 at halt B. Predict the sign of R_A and why.

   Explorer: $R_A =$ _____ kgf $R_B =$ _____ kgf

4. **Find the critical position.** With the same loads, which load should stand on C for the largest R_B: the 3 kg or the 2 kg? Prediction and reason:

   Explorer: largest $R_B =$ _____ kgf with P_1 at halt _____. Was your prediction right? _____

5. **Use the areas.** Show only R_E and shade the areas. Predict the sign of R_E under a uniform load over the whole model.

   Explorer: net area _____ mm, so for $w = 0.015\,\text{kg/mm}$, $R_E =$ _____ kgf.

In the laboratory: after Test 1, open the *Lab* tab of the same page and type your gauge-A readings into the Test 1 table. Note the largest absolute difference from theory and at which halt it occurs: _____ at halt _____. Is it within the ± 0.03 expected from a ± 5 mm placement error and ± 1 division? _____